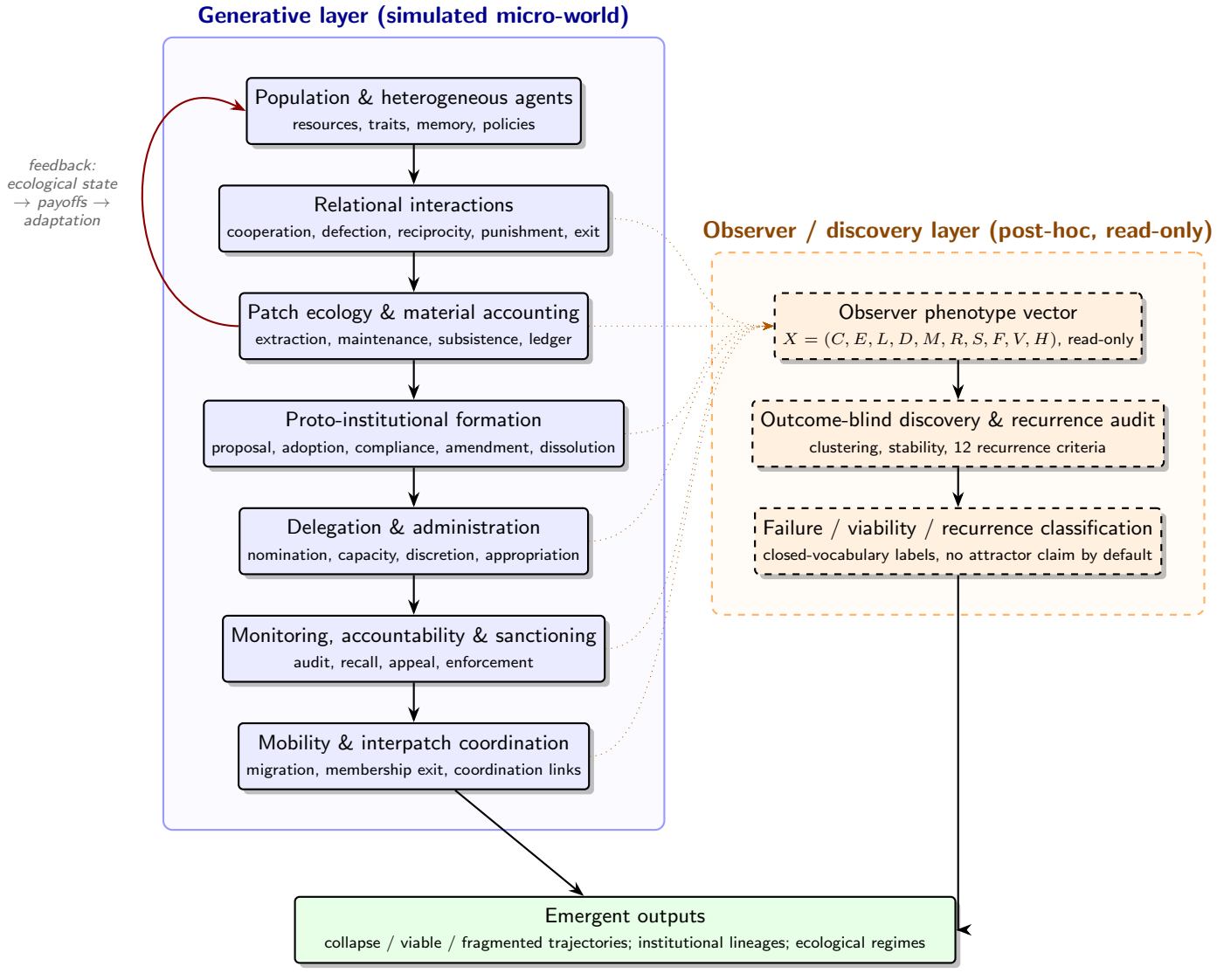


Solid arrows: generative causal sequencing within one simulation step. Dotted arrows: read-only observation into the observer layer (never written back). Dashed boxes: post-hoc analysis, never a generative input.